

Prerequisites

Note
Complete [Lab 2 - Creating a Basic Web Application Using Kiro CLI](#)

Important
Use Kiro **CLI** for this section

Step 1: Understanding Advanced Prompting Techniques

Before we start enhancing our application, let's understand some principles of effective prompting with Kiro:

- 1. **Be specific and detailed:** Provide context, requirements, and constraints
- 2. **Break down complex tasks:** Ask for one feature at a time
- 3. **Iterative refinement:** Start with a basic implementation, then ask for improvements
- 4. **Provide examples:** When possible, show examples of what you want
- 5. **Ask for explanations:** Request explanations of how the code works

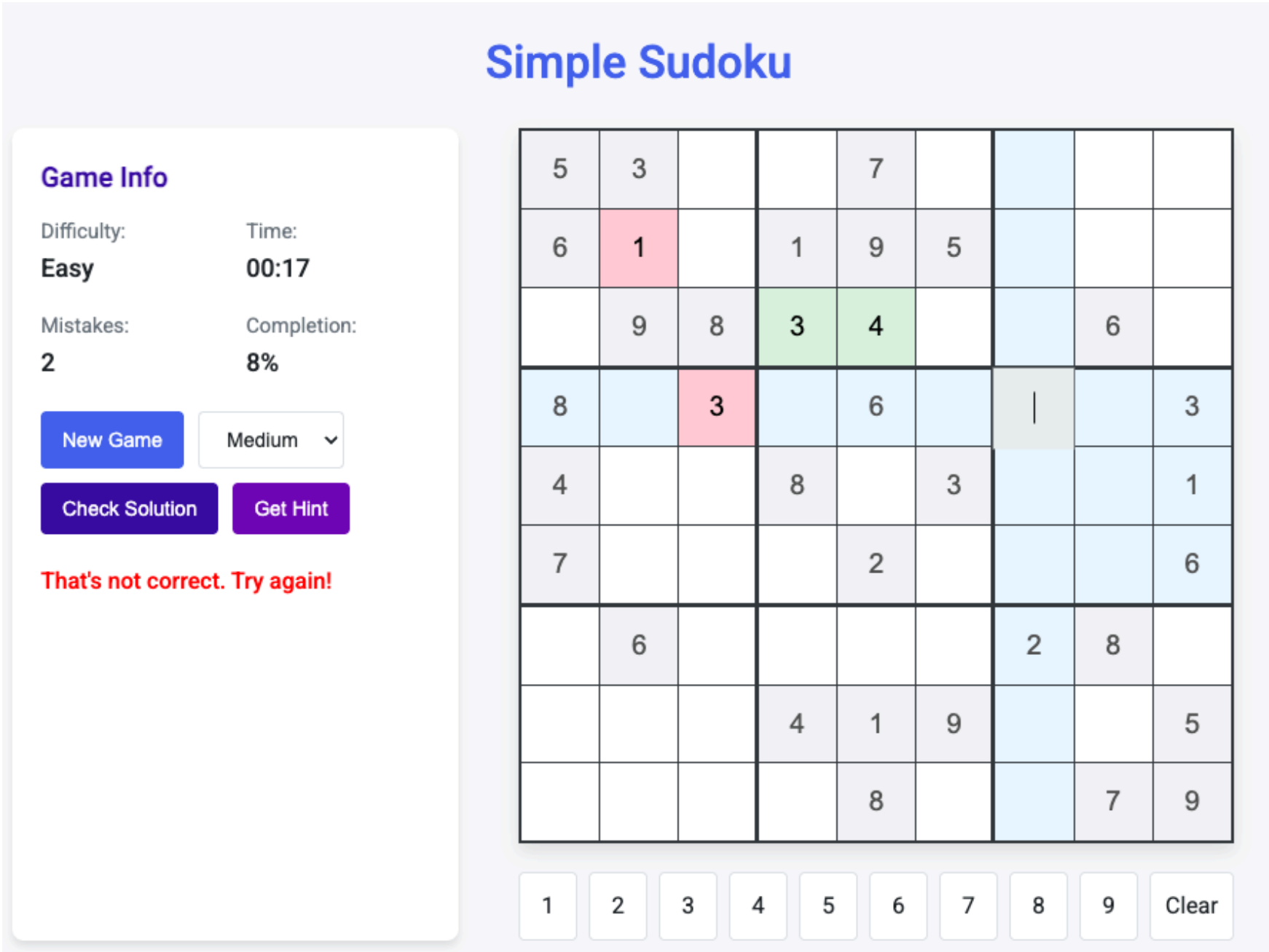
Step 2: Enhancing the UI with Detailed Prompts

- 1. Use a detailed prompt to enhance the UI (IDE or CLI):

```
1 I want to enhance my Sudoku game UI with the following features:
2 1. A modern, clean design with a color scheme that reduces eye strain
3 2. A responsive layout that works well on both desktop and mobile devices
4 3. Verification of grid layout display.
5 4. Visual feedback for selected cells, valid/invalid entries, and completed rows/columns/boxes
6 5. A game info panel showing:
7 - Current difficulty level
8 - Timer
9 - Number of mistakes
10 - Completion percentage
11 6. Animations for cell selection and number placement
12 7. Support for both mouse and keyboard input
```



- 2. Review the result



(Optional) Step 3: Grid layout issue???

After implementing the enhanced features, you can use Kiro to fix the UI issue such as grid layout:

```
1 I found grid layout does not fit with playing box. please fix
```



(Optional) Step 4: Refining and Optimizing with Iterative Prompts

After implementing the enhanced features, you can use Kiro to help refine and optimize your code:

- 1. For performance optimization:

```
1 I've implemented the enhanced Sudoku game, but I'm concerned about performance, especially:
2
3 1. The solver algorithm seems slow on difficult puzzles
4 2. There's a noticeable delay when validating the board after each input
5 3. The UI becomes less responsive when using the visualization feature
6
7 Please suggest optimizations to improve performance while maintaining functionality.
```



- 2. For code quality and organization:

```
1 I want to improve the organization and maintainability of my Sudoku game code:
2
3 Current issues:
4 - The code has grown large and is becoming difficult to manage
5 - There's some duplication in the validation and generation logic
6 - Event handlers are getting complex with multiple responsibilities
7
8 Please suggest how I can:
9 1. Refactor the code using appropriate design patterns
10 2. Separate concerns (UI, game logic, state management)
11 3. Improve code readability and maintainability
12 4. Add proper error handling
```



(Optional) Step 5: Testing Your Enhanced Application

- 1. Test all the new features you've implemented
- 2. If you encounter any issues, use Kiro to help debug:

```
1 "I'm having an issue with [specific feature]. Here's the error I'm seeing: [error details].
2 Here's the relevant code: [paste code]. How can I fix this?"
```



Next Steps

Congratulations! You've enhanced your Sudoku web application using advanced prompting techniques with Kiro. Your application now has more sophisticated features and a better user experience.

In the next lab, you'll learn how to deploy your application to AWS using CDK.